

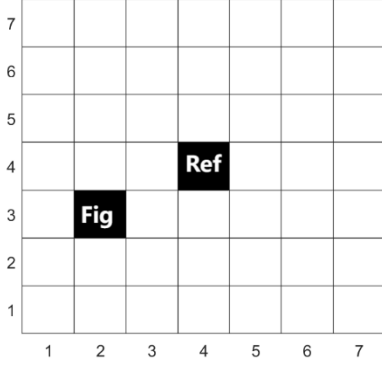


Figure 1: The 7*7 grid plane in spatial representation tasks. The cells noted as 'Ref' and 'Fig' represent the reference object and the figure object respectively, the former of which is always located at the center ([4,4]) while the latter might appear in all the other 48 cells ([2,3] for instance).

Spatial Rating Task. To address the limitation that some LLMs provide only general and coarse terms instead of detailed spatial prepositions in the spatial generation task, a spatial rating task was introduced to further examine their spatial cognition. Unlike the spatial generation task, which required free-form responses, the spatial rating task presented LLMs with predefined statements about the locations of two objects. The models were then required to rate the applicability of these spatial statements on a scale from 1 to 7, where 1 indicated "least appropriate" and 7 indicated "most appropriate." Two reference-figure object pairs—"computer-ring" and "apple-fish"—were selected for this task, combined with four types of spatial relationships: "above," "below," "left," and "right." This design resulted in 384 trials (48 locations * 2 object pairs * 4 relationships). The query prompt for this task followed a specific template, where placeholders were replaced with appropriate values for each trial. The complete set of prompts is available in the supplementary material B.

3.2 LLMs evaluated

The LLMs evaluated in this study include both open-source and closed-source models, incorporating several SOTA models: GPT-3.5-Turbo, GPT-4 (via Azure OpenAI API), Qwen-Turbo, ZhipuAI, and Llama3-8B. To explore the effect of model output variability, experiments were conducted across three temperature settings (0, 0.5, 1) for each LLM. Temperature is a key parameter that controls the uncertainty in the generated content. A higher tem-

perature encourages more diverse and creative responses, but may also reduce reliability and precision. Since this study aims to assess both the creativity and accuracy of LLMs in generating spatial prepositions to describe spatial relationships, varying the temperature allowed for a comprehensive evaluation of the models' ability to balance creativity with precision. Consequently, the spatial representation tasks were repeated across these different temperature settings to account for variability in the models' responses.

3.3 Baseline and evaluation metrics

According to previous studies, most spatial terms used by humans to describe spatial relationships can be categorized into two main types: horizontally oriented and vertically oriented prepositions (Munnich et al., 2001; Hayward and Tarr, 1995). Specifically, horizontally oriented prepositions (e.g., "above" and "below") describe the position of the figure object relative to the reference object in terms of horizontal relations, while vertically oriented prepositions (e.g., "left" and "right") capture vertical relationships between the two objects.

For the spatial generation task, the proportion of horizontally and vertically oriented spatial prepositions used in the LLMs' responses was computed for each cell in the 7*7 grid, with averages taken across different scenarios. Since the concept of 'front' or 'behind' does not apply on a 2D plane, responses involving such prepositions were considered nonsensical or ineffective. Additionally, as neither angles nor compass directions were allowed in the prompts to LLMs, the models' adherence to the instructions was evaluated by examining the proportion of invalid responses. Given that LLMs often use both horizontal and vertical spatial terms simultaneously when describing spatial relationships, the first spatial preposition that appeared in the models' responses was taken as the primary indicator of their axial preference.

In the spatial rating task, LLMs' ratings of statements regarding the spatial relationships between the figure object and the reference object were averaged across all scenarios for each location. To better understand LLMs' basic spatial perception, the 7*7 grid was divided into four 3*7 sub-grids (up, down, left, and right relative to the centrally positioned reference object at [4,4]). The ratings for each sub-grid were then compared to those from the other three sub-grids. This analysis aimed to

Temperature	0	0.5	1
GPT-4	91.25%	94.17%	88.75%
GPT-3.5-Turbo	40.83%	43.33%	39.58%
Qwen-Turbo	71.67%	65.83%	54.17%
ZhipuAI	95.42%	94.17%	93.33%
Llama3-8B	28.75%	25.83%	30.42%

Table 1: Validness of LLMs’ responses on the spatial generation task.

reflect the models’ ability to recognize and distinguish primary axial relations.

In both spatial tasks, LLMs’ performance was compared to that of humans based on a previous related study (Hayward and Tarr, 1995). Specifically, the Euclidean distance between the rating matrices of LLMs and humans was calculated and normalized to quantify the difference in performance. The relative difference, denoted as $Diff_{norm}$, is formulated as follows. A smaller value of $Diff_{norm}$ indicates a closer match between the performance of the models and humans.

$$Diff_{norm} = \frac{\|LLM_{matrix} - Human_{matrix}\|_F}{\max(\|LLM_{matrix}\|_F, \|Human_{matrix}\|_F)}$$

(F means Frobenius norm; *matrix* denotes proportion or mean rating.)

4 Results

4.1 Spatial representations of LLMs are directionally imbalanced and vertically more efficient

The spatial prepositions selected by LLMs to describe the spatial relationships between the figure object and the reference object exhibit considerable diversity, particularly in more advanced models. Horizontally oriented spatial terms include "left", "right", "beside", and "next to", while vertically oriented terms encompass "above", "below", "up", "low(er)", "ahead", and "beyond". In addition to these axial prepositions, LLMs’ responses also contain some non-axial spatial terms, such as "diagonal", "southwest", "behind", and "near". These non-axial terms, though less frequent, are considered inappropriate as they do not adhere to the instructions specifying axial relationships in a 2D grid. Responses incorporating these terms were therefore coded as invalid.

The proportions of invalid responses from the five LLMs under three different temperature settings are presented in Table 1. This data reveals

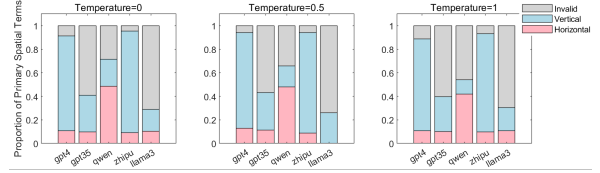
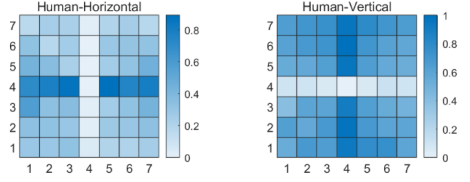


Figure 2: Proportions of different types spatial prepositions shown in models’ responses at first. GPT-4 and ZhipuAI show better validness. Most LLMs except Qwen-Turbo tend to prefer vertically oriented spatial terms relative to horizontally oriented spatial terms.

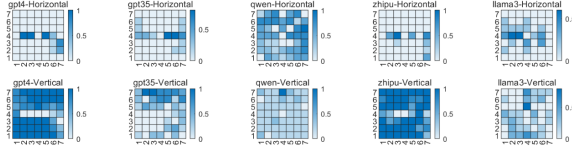
that SOTA models such as GPT-4 and ZhipuAI consistently provide more accurate and effective spatial representations, more closely aligning with human-like spatial reasoning. These models also demonstrate a preference for describing spatial relationships along axial directions. Moreover, vertically oriented prepositions are more frequently chosen as the primary descriptors, a trend also observed in human spatial language. The proportions of three types of spatial prepositions (horizontal, vertical, and others) in LLMs’ responses across varying temperature levels are shown in Fig. 2. The results suggest that temperature settings only have a subtle effect on the models’ performance in the spatial generation task. Notably, most models, with the exception of Qwen-Turbo, tend to use vertically oriented spatial prepositions as their primary means of describing spatial relationships between objects on a 2D plane.

4.2 Resemblance of LLMs to humans in preference of vertical spatial terms

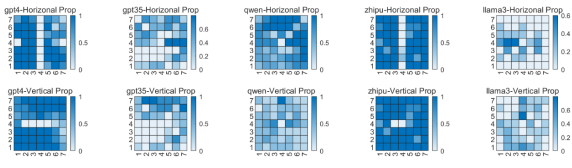
The proportions of horizontally and vertically oriented spatial prepositions that appeared first in the models’ responses at each location are compared with human performance, as derived from the previous study (Hayward and Tarr, 1995). As shown in Fig. 3(a), humans exhibit a clear axial preference when describing spatial relationships. Specifically, horizontally or vertically oriented spatial prepositions are more likely to be chosen as the primary descriptors when the figure object is positioned near the corresponding axis. However, the patterns in the LLMs’ responses to spatial term generation exhibit notable differences (Fig. 3(b)). All models accurately generate horizontal spatial prepositions along the x-axis centered on the reference object, except for Qwen-Turbo. The horizontal prepositions produced by Qwen-Turbo are scattered and lack a clear, consistent pattern.



(a) Humans' choice of each type of spatial terms.



(b) Performance of five LLMs on spatial preposition preference at all cells except the center.



(c) Distribution of horizontal and vertical spatial prepositions appeared in LLMs' responses in the spatial generation task.

Figure 3: Primacy of horizontal and vertical spatial prepositions in LLMs' responses at each location on the 7*7 grids. Considering the subtle influence of temperature on LLMs' generation performance, the temperature underlying results displayed here is 0, whereas results of the other two situations (i.e. 0.5 and 1) are available in Appendix Fig.S1 and S2.

On the other hand, both GPT-4 and ZhipuAI appear to overemphasize encoding spatial relationships in the vertical direction, as they generate a notably higher proportion of vertical spatial prepositions compared to other models. GPT-3.5-Turbo, on the other hand, tends to produce more vertical prepositions when the figure object is located above the reference object. In contrast, Qwen-Turbo still exhibits no discernible pattern in the distribution of vertical spatial prepositions. Llama3-8B, however, demonstrates a clear axial effect, with consistent performance in both vertical and horizontal directions.

When considering the frequency of horizontal and vertical spatial terms combined in the models' responses—without focusing on their primacy—results show that GPT-4 and ZhipuAI encode both horizontal and vertical relationships comprehensively (Fig. 3(c)). These models provide a dense representation, employing spatial terms in both directions across nearly every position. Llama3-8B's performance mirrors the findings in

Temperature	0	0.5	1
GPT-4	0.787	0.785	0.711
GPT-3.5-Turbo	0.686	0.722	0.713
Qwen-Turbo	0.579	0.547	0.570
ZhipuAI	0.770	0.776	0.763
Llama3-8B	0.775	1	0.766

Table 2: Horizontal difference between the performance of LLMs and humans.

Temperature	0	0.5	1
GPT-4	0.331	0.378	0.311
GPT-3.5-Turbo	0.662	0.632	0.695
Qwen-Turbo	0.689	0.754	0.843
ZhipuAI	0.360	0.346	0.352
Llama3-8B	0.778	0.774	0.764

Table 3: Vertical difference between the performance of LLMs and humans.

the primacy analysis discussed earlier. In contrast, no clear pattern emerges in the responses of GPT-3.5-Turbo and Qwen-Turbo.

The disparity between the performance of LLMs and humans in the spatial generation task is further computed and presented in Table 2 (for horizontal directions) and Table 3 (for vertical directions). In terms of human-like performance, the spatial representations of both GPT-4 and ZhipuAI are generally more similar to humans in the vertical direction, as their normalized difference ($Diff_{norm}$ index) is lower than 0.5, outperforming all other models. However, in the horizontal direction, the normalized difference between all models and humans exceeds 0.5, regardless of the temperature setting. Therefore, only SOTA models like GPT-4 resemble humans in choosing vertically oriented spatial prepositions to characterize spatial relationships.

4.3 SOTA LLMs demonstrate a deficiency in representing rightward spatial relationships

To gain a more nuanced understanding of LLMs' spatial representation, models were tasked with rating the applicability of statements describing four types of spatial relations between the reference object and the figure objects. A comparison was made between the average ratings of spatial statements describing relations where the figure objects are located in the corresponding subgrid area (e.g., the "above" relation used for figure objects in the upper